

Subject Name	BTL1123/BCL1123 Internet of Things
Assignment Title	Final Report & Video
Semester	May-Aug 2026
Release Date	As in LMS
Due Dates	As in LMS
Total Marks	100%

Final Assessment: Project Development and Report

IMPORTANT INFORMATION

You are required to submit your work within the bounds of the University Infringement of Assessment Regulations. Plagiarism, paraphrasing and downloading large amounts of information from external sources will not be tolerated and will be dealt with severely. However, you should make full use of any source material, which would normally be an occasional sentence and/or paragraph (duly referenced) followed by your own critical analysis/evaluation.

The work must be entirely your own. You will receive ZERO (0) mark for work that is not your own. The safety of your assessments is your responsibility. You must not permit another student access to your work. Where referencing is required, APA referencing system must be used.

Learning Outcome

Develop simple smart IoT solutions using relevant programming language. [CLO 3]

Project Overview

This is an **individual assessment** carrying a weightage of 100% of the final project marks. You are required to execute the IoT proposal you submitted in the assignment phase. You will use VS Code, Wokwi (for simulation), and Blynk (for dashboard and alerts) as your development and testing environment.

The final project focuses heavily on the technical solution, coding, and simulation for your smart accommodation IoT system.

Task Instructions

- **Simulation Environment:** Implement your system using Wokwi integrated with VS Code. Connect your virtual hardware to Blynk (or a similar relevant platform) for cloud connectivity and dashboard visualization.
- **Screenshots & Recording Evidence:** Capture high-quality screenshots & Videos of the proposed solution (circuit layout in Wokwi, dashboard in Blynk, and relevant data flow diagrams).
- **Coding Explanation:** Clearly comment the code used to connect the sensors, process the data, and transmit it to the cloud.
- **Security Components:** Discuss and explain the security components considered in your system (e.g., Wi-Fi authentication, secure MQTT, token protection).
- **Uniqueness:** Highlight the special uniqueness of your implemented system in the report.
- **Video Presentation:** Record a presentation video (MAXIMUM 15 minutes) to demonstrate the entire project running in real-time. **Ensure your camera is turned on and your face is visible throughout video. Marks will be deducted if the camera is off during the presentation.**

Final Report Format

Your final report must include the following sections:

1. **Cover Page**
 2. **Executive Summary:** A brief summary of the final project.
 3. **Implementation Explanations:** Detailed breakdown using screenshots, pictures, diagrams, coding snippets, and flowcharts to elaborate the system.
 4. **Security Components:** Explanation of the security measures considered in the system.
 5. **Reflections:** Reflections on the project, challenges faced during development, and possible solutions to overcome them.
 6. **Summary**
 7. **References:** Include citations for any source of references used (APA format).
 8. **URL Link to the share folder to access all components of the final project.**
- Make sure all information required is accurate. All information retrieved from the Internet or textbook must be cited accordingly.
 1. Reference from Wikipedia is not allowed.
 2. Create a clean, logically structured Table of Contents for the report, reflecting the sections in the task instructions.

FORMATTING INSTRUCTIONS

- Font
 - Use Times New Roman font, 12-point font style
 - Use bold, italics, and underline in an appropriate manner
- Paragraph
 - Alignment Justified
 - Indentation First line
 - Use 1.5 line spacing
- Styles
 - Use headings and subheadings.
- Page layout
 - Orientation portrait and landscape
 - Page break and section break
 - Different Header and footer for different sections
- Picture tools
 1. Format picture
 2. Insert caption
- Use APA referencing and citations.

Submission Instructions

- Submission includes:
 - Report (DOCX)
 - Video (MP4)
 - entire VS project folder
 - a readme.txt.
- Upload all the above to a shared folder
- Ensure the shared folder link is accessible. It is the student's responsibility to make sure the marker can view it.
- Note: If the marker cannot open any of the files due to restricted access permissions or unavailability, it will be treated as missing component and a grade of 0 marks will be given for the missing component.
- The readme.txt **MUST** contain your Blynk (or relevant platform) login credentials so the examiner can log in and test your project.
- Submit the share link the LMS system before the deadline.
- The work must be entirely your own. Failure to provide working code or credentials will result in a loss of marks.
- Submission Deadline and Late Policy:
 - Please submit your assignment by or before the deadline.
 - **LATE SUBMISSION** will be **DEDUCTED 10% penalty A DAY** up to a **maximum of FIVE (5) days**, were subsequently, the assignment will be awarded **ZERO (0)** mark.
 - **ZERO (0)** mark will be given if there is any form of copying/plagiarism.
 - **ZERO (0)** mark will be given if you fail to submit your work or fail to submit all the necessary files.

Final Project Marking Scheme

Final Project Question Summary

Students are required to individually execute their IoT proposal using VS Code, Wokwi, and Blynk. They must simulate the circuit, connect it to the cloud, configure a dashboard, and explain the coding and security considerations. Submissions include a report with screenshots, the VS project code, login credentials, and a 15-minute demonstration video.

Marking Scheme

Your work will be assessed based on the following criteria.

- **System Integration & Simulation (20%):** Efficient hardware integration in Wokwi with a stable simulation.
- **Cloud Connectivity & Data Transmission (15%):** Reliable real-time data transmission and code logic for managing connections.
- **Dashboard Visualization & Interface (15%):** Clear, intuitive dashboard configuring trends and alerts effectively (e.g., via Blynk).
- **Security Considerations (10%):** Clear and technically accurate explanation of multiple security layers considered for your system.
- **Report (Implementation & Reflections) (20%):** High-quality documentation, code explanations, and thoughtful reflections on challenges.
- **Video Presentation & Demonstration (20%):** Smooth, real-time demonstration validating the system logic within 15 minutes.